

(A1)

$$a. \frac{1}{4} \quad b. \frac{1}{3} \quad c. \frac{6}{16} \quad d. \frac{4}{7} \quad e. \frac{7}{8}$$

(A2)

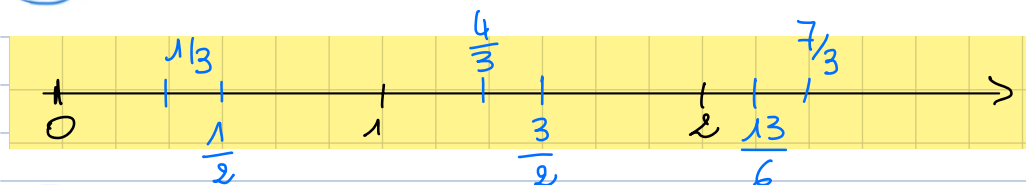
$$a. \frac{9}{4} = 2 + \frac{1}{4} \quad b. \frac{5}{2} = 2 + \frac{1}{2}$$

(A3)

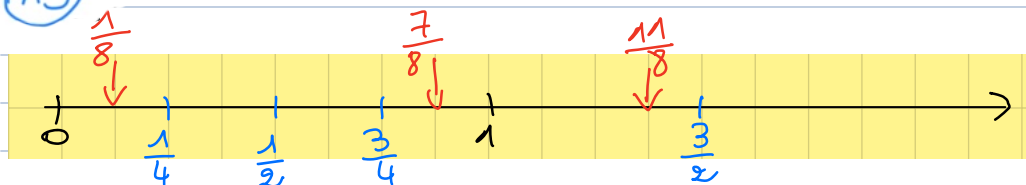
$$a. \frac{9}{6} \quad b. \frac{8}{5}$$

$$c. \frac{13}{11} \quad d. \frac{11}{13}$$

(A4)



(A5)



(B1)

$$a. \frac{4}{5} = \frac{4 \times 3}{5 \times 3} = \frac{12}{15}$$

$$b. \frac{5}{9} = \frac{5 \times 4}{9 \times 4} = \frac{20}{36}$$

$$c. \frac{0.5}{2.6} = \frac{0.5 \times 10}{2.6 \times 10} = \frac{5}{26}$$

$$d. \frac{45}{25} = \frac{9 \times 5}{5 \times 5} = \frac{9}{5}$$

(B2)

$$a. \frac{6}{8} = \frac{6 \div 2}{8 \div 2} = \frac{3}{4}$$

$$b. \frac{10}{25} = \frac{10 \div 5}{25 \div 5} = \frac{2}{5}$$

$$c. \frac{9}{12} = \frac{9 \div 3}{12 \div 3} = \frac{3}{4}$$

$$d. \frac{42}{14} = \frac{42 \div 14}{14 \div 14} = \frac{3}{1} = 3$$

(B3)

$$a. \frac{8}{5} = \frac{16}{10} = \frac{32}{20} = \frac{64}{40} = \frac{128}{80} = \frac{256}{160} = \frac{512}{320} = \frac{1024}{640} = \frac{2048}{1280} = \frac{4096}{2560} = \frac{8192}{5120} = \frac{16384}{10240} = \frac{32768}{20480} = \frac{65536}{40960} = \frac{131072}{81920} = \frac{262144}{163840} = \frac{524288}{327680} = \frac{1048576}{655360} = \frac{2097152}{1310720} = \frac{4194304}{2621440} = \frac{8388608}{5242880} = \frac{16777216}{10485760} = \frac{33554432}{20971520} = \frac{67108864}{41943040} = \frac{134217728}{83886080} = \frac{268435456}{167772160} = \frac{536870912}{335544320} = \frac{1073741824}{671088640} = \frac{2147483648}{1342177280} = \frac{4294967296}{2684354560} = \frac{8589934592}{5368709120} = \frac{17179869184}{10737418240} = \frac{34359738368}{21474836480} = \frac{68719476736}{42949672960} = \frac{137438953472}{85899345920} = \frac{274877906944}{171798691840} = \frac{549755813888}{343597383680} = \frac{1099511627776}{687194767360} = 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C]

25 p 42

a. $\frac{3}{5} > \frac{3}{7}$

b. $\frac{7}{3} < \frac{7}{2}$

26 p 42

a. $\frac{5}{9} < \frac{5}{3}$

b. $\frac{6}{7} > \frac{6}{8}$

27 p 42

a. $\frac{3}{4} < 1 - \frac{7}{6} > 1$ donc $\frac{3}{4} < \frac{7}{6}$

6 p 37

a. $\frac{4}{7} < 1$ et $\frac{5}{4} > 1$ donc $\frac{4}{7} < \frac{5}{4}$

7 p 37

a. $\frac{4}{7} > \frac{1}{2}$ et $\frac{1}{4} < \frac{1}{2}$ donc $\frac{1}{4} < \frac{7}{4}$

b. $\frac{11}{8} > 1$ et $\frac{7}{12} < 1$ donc $\frac{8}{11} > \frac{7}{12}$

c. $\frac{3}{5} < 1$ et $\frac{11}{9} > 1$ donc $\frac{3}{5} < \frac{11}{9}$

b. $\frac{8}{11} > \frac{1}{2}$ et $\frac{11}{23} < \frac{1}{2}$ donc $\frac{6}{11} > \frac{11}{23}$

c. $\frac{5}{9} > 1$ et $\frac{5}{4} < 1$ donc $\frac{5}{9} > \frac{5}{4}$

d. $\frac{11}{12} < 1$ et $\frac{12}{11} > 1$ donc $\frac{11}{12} < \frac{12}{11}$

c. $\frac{3}{5} > \frac{1}{2}$ et $\frac{2}{7} < \frac{1}{2}$ donc $\frac{3}{5} > \frac{2}{7}$

28 p 42

a. $\frac{2}{3} < \frac{5}{6}$

b. $\frac{13}{5} < \frac{31}{10}$

c. $\frac{3}{4} > \frac{7}{12}$

d. $\frac{6}{8} > \frac{4}{12}$

D]

30 p 42

a. $\frac{3}{2} = \frac{9}{6}$

b. $\frac{9}{6} + \frac{15}{6} = \frac{24}{6}$

b. $\frac{3}{4} = \frac{6}{8}$

c. $\frac{11}{8} - \frac{6}{8} = \frac{5}{8}$

c. $\frac{2}{3} = \frac{6}{9}$

d. $\frac{7}{9} - \frac{6}{9} = \frac{1}{9}$

d. $\frac{2}{5} = \frac{4}{10}$

e. $\frac{7}{10} - \frac{4}{10} = \frac{3}{10}$

31 p 42

a. $\frac{3}{2} = \frac{6}{4}$

b. $\frac{5}{4} + \frac{6}{4} = \frac{11}{4}$

b. $\frac{2}{3} = \frac{6}{9}$

c. $\frac{11}{9} - \frac{6}{9} = \frac{5}{9}$

33 p 42

1. 6

a. $\frac{2}{3} = \frac{4}{6}$; $\frac{1}{2} = \frac{3}{6}$

c. $\frac{5}{4} = \frac{15}{12}$

d. $\frac{15}{12} - \frac{3}{12} = \frac{12}{12} = 1$

d. $\frac{1}{2} = \frac{5}{10}$

e. $\frac{7}{10} - \frac{5}{10} = \frac{2}{10} = \frac{1}{5}$

2. a. $\frac{4}{6} + \frac{3}{6} = \frac{7}{6}$

b. $\frac{4}{6} - \frac{3}{6} = \frac{1}{6}$

34 p 42

1. 12

a. $\frac{2}{3} = \frac{8}{12}$

b. $\frac{3}{4} = \frac{9}{12}$

8 p 37

a. $3 = \frac{3}{1} = \frac{15}{5}$

c. $\frac{15}{5} - \frac{4}{5} = \frac{11}{5}$

2. a. $\frac{8}{12} + \frac{9}{12} = \frac{17}{12}$

b. $\frac{9}{12} - \frac{8}{12} = \frac{1}{12}$

b. $4 = \frac{4}{1} = \frac{32}{8}$

c. $\frac{32}{8} + \frac{7}{8} = \frac{39}{8}$

c. $5 = \frac{5}{1} = \frac{15}{3}$

d. $\frac{15}{3} + \frac{2}{3} = \frac{17}{3}$

9 p 37

a. multiple de 5 et 3: 15

a. $\frac{5}{3} = \frac{25}{15}$

b. $\frac{4}{5} = \frac{12}{15}$

c. $\frac{25}{15} - \frac{12}{15} = \frac{13}{15}$

b. multiple de 7 et 8: 56

a. $\frac{3}{7} = \frac{24}{56}$

b. $\frac{5}{8} = \frac{35}{56}$

c. $\frac{24}{56} + \frac{35}{56} = \frac{59}{56}$

c. multiple de 5 et 3: 15

a. $\frac{4}{5} = \frac{12}{15}$

b. $\frac{2}{3} = \frac{10}{15}$

c. $\frac{12}{15} + \frac{10}{15} = \frac{22}{15}$

46 p44

1. b.

2. d.

3. a.

4. c.

47 p44

$$a. \frac{7}{5} = 1 + \frac{2}{5} = 1,4$$

$$b. 2,5 = 2 + \frac{1}{2} = 2,5$$

49 p44

1. c.

2. b. d.

3. a. e.

50 p44 DÉFI

$$\frac{4}{5} (= 0,8)$$

51 p44

$$a. \frac{7}{5} = \frac{56}{40} \quad \text{et} \quad \frac{9}{8} = \frac{45}{40} \quad \text{donc} \quad \frac{7}{5} > \frac{9}{8}$$

$$b. \frac{2}{3} = \frac{16}{24} \quad \text{et} \quad \frac{5}{8} = \frac{15}{24} \quad \text{donc} \quad \frac{2}{3} > \frac{5}{8}$$

$$c. \frac{3}{5} = \frac{12}{20} \quad \text{donc} \quad \frac{11}{5} < \frac{12}{5}$$

$$d. \frac{13}{25} = \frac{26}{50} \quad \text{et} \quad \frac{1}{2} = \frac{25}{50} \quad \text{donc} \quad \frac{13}{25} > \frac{1}{2}$$

53 p44

$$a. \frac{7}{4} = 1,75 < 1,8$$

$$b. \frac{9}{8} = 1,125 > 1,1$$

$$c. \frac{1}{5} = 0,20 \quad \text{donc} \quad 2,25 > 2 + \frac{1}{5} \quad d. 3 + \frac{2}{3} = \frac{9}{3} + \frac{2}{3} = \frac{11}{3}$$

54 p44 *

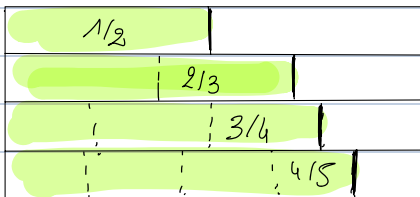
$$25\% < 1 + \frac{3}{5} < \frac{11}{6} < \frac{17}{8} < 225\% < 2,5 < 3 - \frac{2}{5}$$

55 p44 *

$$4 - \frac{1}{5} > 300\% > \frac{19}{8} > 2 + \frac{1}{3} > \frac{25}{12} > 208\% > 1,75$$

57 p44 DÉFI

a.



b. étape 100 : 1 le dénominateur à l'étape 101 est plus grand qu'à l'étape 100 et

étape 101 : 1 les fractions ont le même numérateur donc $\frac{1}{101} > \frac{1}{102}$

58 p45

$$a. \frac{3}{4} + \frac{5}{8} = \frac{11}{8}$$

$$b. \frac{5}{7} + \frac{3}{21} = \frac{18}{21}$$

$$c. \frac{9}{10} - \frac{2}{5} = \frac{5}{10} = \frac{1}{2}$$

$$d. \frac{5}{8} - \frac{1}{2} = \frac{1}{8}$$

$$e. \frac{5}{12} + \frac{5}{6} = \frac{10}{12} = \frac{5}{6}$$

$$f. \frac{3}{4} + \frac{1}{5} + \frac{7}{10} = \frac{33}{20}$$

59 p45 SANS CALCULATRICE et simplifier

$$a. \frac{4}{12} + \frac{6}{18} = \frac{24}{36} = \frac{2}{3}$$

$$b. \frac{10}{14} - \frac{6}{21} = \frac{3}{7}$$

$$c. 2 - \frac{3}{12} = \frac{21}{12} = \frac{7}{4}$$

$$d. \frac{21}{15} + \frac{12}{10} = \frac{66}{30} = \frac{11}{5}$$

$$e. \frac{15}{6} + 4 = \frac{39}{6}$$

$$f. \frac{10}{15} - \frac{1}{9} = 0$$

$$a. \text{bleu} = \frac{3}{4}$$

$$\text{rouge} = \frac{2}{4} = \frac{1}{2}$$

$$\text{orange} = 1$$

$$\text{vert} = \frac{5}{8}$$

61 p45

$$a. \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$b. \frac{2}{3} + \frac{9}{12} = \frac{17}{12}$$

$$c. \frac{3}{10} + \frac{2}{5} = \frac{7}{10}$$

$$d. \frac{2}{6} + \frac{1}{2} = \frac{5}{6}$$

83 p48 *

$$b. \frac{3}{4} + \frac{1}{2} + 1 = \frac{9}{4}$$

$$1 + \frac{5}{8} - \frac{1}{2} = \frac{9}{8}$$

$$\frac{1}{2} + \frac{3}{6} - 1 = \frac{1}{4}$$

$$1 + \frac{5}{8} - \frac{3}{4} = \frac{7}{8}$$

$$9 - 8$$

65 p45 *

$$a. \square = \frac{3}{6} = \frac{1}{2} \quad \text{donc} \quad \square + \frac{3}{8} = \frac{7}{8}$$

$$b. \bigcirc = \frac{7}{12} \quad \text{donc} \quad \bigcirc - \frac{1}{6} = \frac{5}{12}$$

$$c. \diamond = \frac{7}{20} \quad \text{donc} \quad \diamond = \frac{4}{60} = \frac{1}{15}$$

87 p49

$$a. \frac{1}{2} + \frac{3}{4} - \frac{2}{3} = \frac{7}{12}$$

$$b. \frac{9}{12} + \frac{1}{12} = \frac{10}{12}$$

